Variational and Optimization-based Approaches for Event-Based Vision Applications

From unobserved modes to well-posed estimators

Daniel Blauvelt

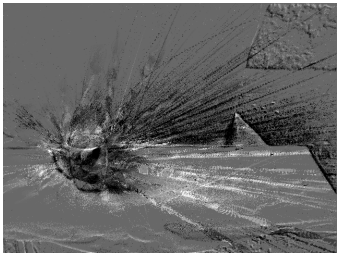
Department of Mathematics ▪ George Mason University
Dissertation Director: Harbir Antil

Dissertation Defense ▪ Fall 2026

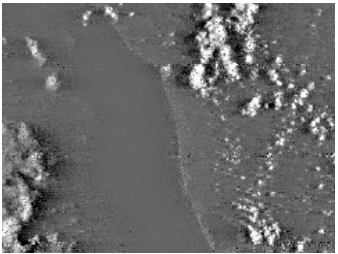
Events alone recover a handheld office, a bullet hitting a cup, and Earth from orbit



Office, handheld



Bullet hits a cup

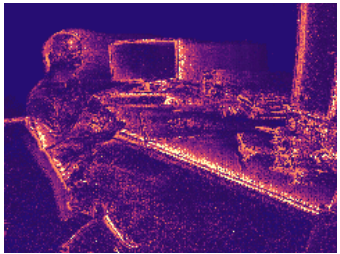


Earth from the ISS (Falcon Neuro)

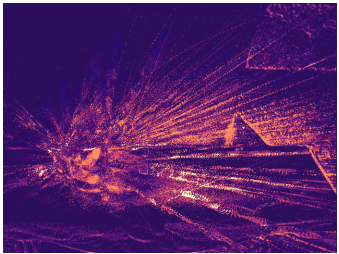
What can brightness changes reveal about images, motion, and depth—and at what computational cost?

Dissertation: Figure 2.7, Figure 2.8, and Figure 2.9; Section 2.3.3. Data: office [36], cup [37], Falcon Neuro [38].

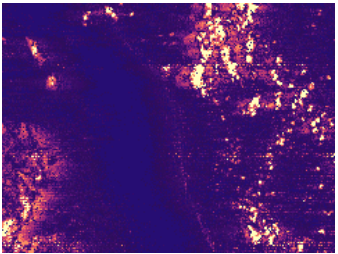
Events alone recover a handheld office, a bullet hitting a cup, and Earth from orbit



Office, handheld



Bullet hits a cup

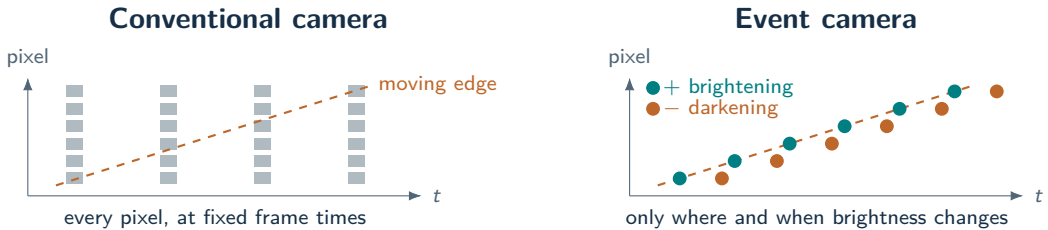


Earth from the ISS (Falcon Neuro)

What can brightness changes reveal about images, motion, and depth—and at what computational cost?

Dissertation: Figure 2.7, Figure 2.8, and Figure 2.9, false-color renderings of the same reconstructed frames.

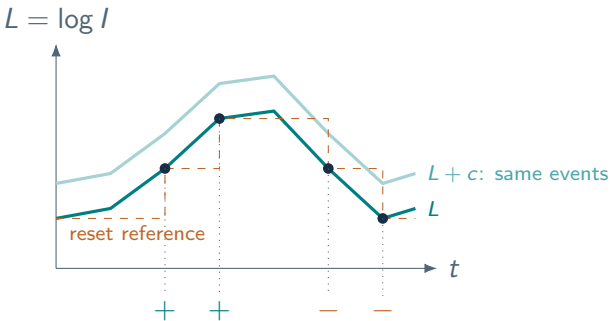
Why an event camera? Each pixel reports its own changes, with microsecond timing and 140 dB range



- Microsecond timestamps, about 140 dB dynamic range versus 60 dB, low power, little motion blur [1].

Dissertation: Chapter 2, neuromorphic camera overview; [1]. Diagram is schematic.

Each event marks one threshold of log-brightness change, so the baseline level stays free



An event is a tuple

$$e_k = (\mathbf{x}_k, s_k, p_k), \qquad p_k = \pm 1.$$

For the ideal threshold model,

$$L(\mathbf{x}_k, s_k) - L(\mathbf{x}_k, s_k^-) = p_k c_{\mathrm{th}}.$$

Dissertation: Eq. 2.1; Eq. 3.3; ideal threshold model [20]. Diagram is schematic.

From events to images, motion, and depth: each estimate has its own unobserved quantity

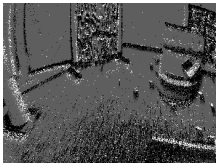
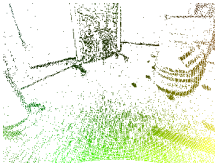


Image reconstruction (Ch. 2)

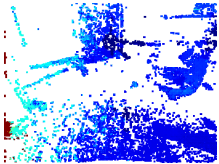
Unobserved: the brightness baseline



Optical flow (Ch. 3)

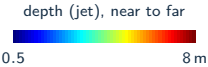
Unobserved: motion along an edge





Stereo depth (Ch. 4)

Unobserved: which points correspond



Data: MVSEC [35].

Dissertation: Methods: Chapter 2, Chapter 3, and Chapter 4; MVSEC [35].

Three estimators, each well posed

	Reconstruction (Ch. 2)	**Optical flow** (Ch. 3)	**Stereo depth** (Ch. 4)
Model supplies	the baseline, through the penalty	the motion along edges, through smoothness	the match, by encouraging two profiles to agree
Guarantee	unique and stable; one tridiagonal solve per pixel	unique when edges span two directions	unique inner solve at each candidate depth, with frozen weights

The common question: which quantities do the events determine, and which does the model supply?

Dissertation: Table 5.1; Theorem 2.2.5, Theorem 3.6.1, Theorem 4.6.3.

Events fix each pixel's brightness increments and leave one constant free

Observation at one pixel

Choose a unit-mass temporal kernel ϕ_{s_k}:

$$e(t) = \sum_{k \text{ at this pixel}} p_k \phi_{s_k}(t),$$

$$y_i = \int_{t_i}^{t_{i+1}} e(t)\, dt.$$

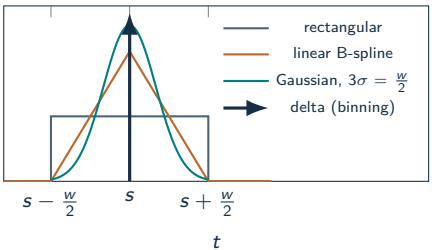
Increments of the unknown history

$$\mathbf{A} = \begin{pmatrix} -1 & 1 & & \\ & -1 & 1 & \\ & & \ddots & \ddots \end{pmatrix},$$

$$\mathbf{A}\mathbf{f} \approx \mathbf{y}, \qquad \ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}.$$

Dissertation: Eq. 2.4, Eq. 2.7, and Lemma 2.2.4.

Each kernel spreads an event's sign over nearby bins and conserves its total



$$y_i = \sum_k p_k \int_{t_i}^{t_{i+1}} \phi_{s_k}(t)\, dt,$$

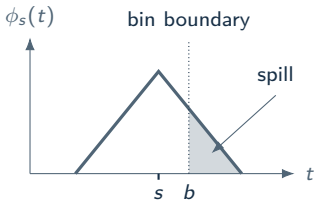
$$\sum_i y_i = \sum_k p_k$$

when every support lies inside $[t_0, t_N]$.

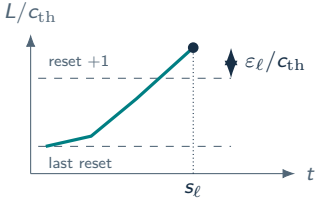
Dissertation: Proposition 2.2.1, Proposition 2.2.2; Table 2.2.

Three discrepancies connect event sums to brightness changes

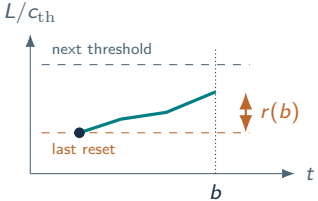
$$\frac{L(b) - L(a)}{c_{\mathrm{th}}} = y(a,b) - \underbrace{B(a,b)}_{\text{kernel spill}} + \underbrace{\frac{1}{c_{\mathrm{th}}} \sum_{a<s_\ell\le b} \varepsilon_\ell}_{\text{overshoot}} + \underbrace{r(b) - r(a)}_{\text{sub-threshold change}} \qquad (a<b)$$



Shaded kernel mass lies beyond b.



Event recorded above threshold.



Change since the last event.

Dissertation: Physical connection (p. 24), Eq. 2.29, arbitrary-endpoint identity; threshold model [20]. Schematic diagram.

A unique, stable estimate for every positive weight

For fixed $\mathbf{\Lambda} = \mathrm{diag}(\lambda_i)$ with every $\lambda_i > 0$,

$$\mathbf{f}^* = \arg\min_{\mathbf{f}\in\mathbb{R}^n} \tfrac{1}{2}\,\|\mathbf{A}\mathbf{f} - \mathbf{y}\|^2 + \tfrac{1}{2}\,\|\mathbf{\Lambda}\mathbf{f}\|^2, \qquad \mathbf{K}_{\mathrm{time}} = \mathbf{A}^\top\mathbf{A} + \mathbf{\Lambda}^\top\mathbf{\Lambda} \succeq \lambda_{\mathrm{min}}^2\mathbf{I}.$$

Unique	$\mathbf{f}^* = \mathbf{K}_{\mathrm{time}}^{-1}\mathbf{A}^\top\mathbf{y}$	one tridiagonal solve per pixel
Stable	$\|\delta\mathbf{f}^*\| \le (2/\lambda_{\mathrm{min}}^2)\,\|\delta\mathbf{y}\|$	one spurious event moves a history by at most $2/\lambda_{\mathrm{min}}^2$
Reference level	$\mathbf{1}^\top\mathbf{\Lambda}^2\mathbf{f}^* = 0$	quiet intervals carry the largest weights and set the level

Dissertation: Eq. 2.8, Theorem 2.2.5, Proposition 2.2.7; Tikhonov regularization [27].

Each pixel costs one tridiagonal sweep; streaming adds constant work per new frame

$$\mathbf{K}_{\text{time}} = \begin{pmatrix} 1+\lambda_1^2 & -1 & & & \\ -1 & 2+\lambda_2^2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 1+\lambda_N^2 & \end{pmatrix}$$

$O(N)$ per pixel	Thomas sweep, no pivoting; tridiagonal and diagonally dominant
Independent pixels	a pixel without events returns $\mathbf{f} = 0$
Newest frame	the forward sweep alone gives it: constant work per pixel for each new interval

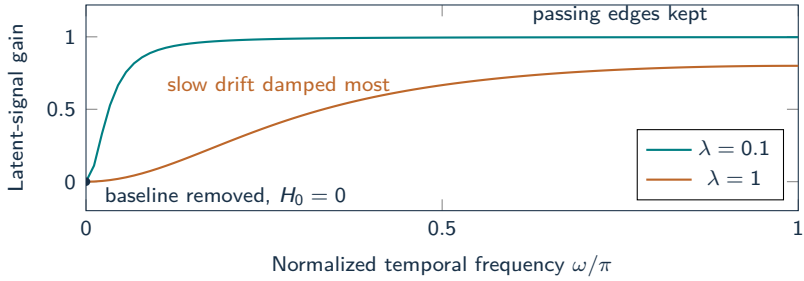
MVSEC indoor flying 1: dense solve 1.5 s → tridiagonal 0.03 s; agrees with MATLAB within 10^{-15}.

Dissertation: Eq. 2.7, Theorem 2.2.5; Thomas algorithm [32]; validation: edge implementation, October 2026.

The penalty removes the baseline and damps slow drift most; passing edges remain

For scalar $\mathbf{\Lambda} = \lambda \mathbf{I}$ and noiseless data $\mathbf{y} = \mathbf{A}\mathbf{f}_{\mathrm{true}}$:

$$g_k = 4 \sin^2 \frac{\pi k}{2n}, \qquad H_k = \frac{g_k}{g_k + \lambda^2}, \qquad H_0 = 0.$$



Dissertation: Lemma 4.4.1, Theorem 4.6.21; cosine eigenbasis [80]. Curves: continuous extension.

Events alone recover scene structure with one convex solve and a unique global minimizer

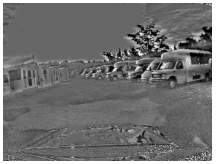
MVSEC [35]
outdoor day 1



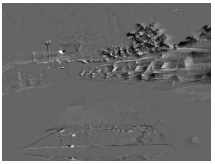
Reference frame



Bilevel v [2]



Bilevel u



Events

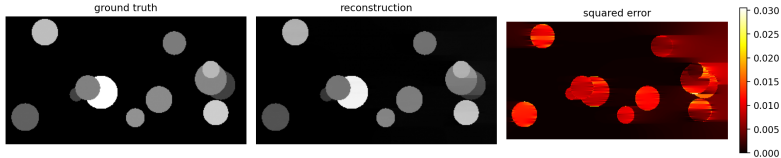
Inputs	events + frames	events
PSNR (dB) indoor / outdoor	40.25 / 36.57 (on its fitted frames)	8.00 / 7.19
Solve	outer + inner	one convex
Guarantee	existence; conditional local uniqueness	unique global minimizer

Dissertation: Section 2.3.2. Antil–Sayre [2], Sec. 3, Eqs. (3.1)–(3.3), Sec. 4.1.4.

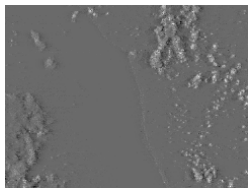
Known motion lets every view constrain one scene

Model $B_{\mathrm{SAR}} = (A \otimes I)\, P$ — θ scene · P known motion · A differences between views

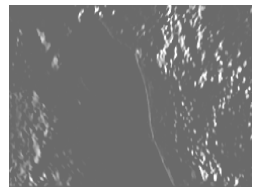
Simulated, known truth (v2e [45]); 1 px shift per view



Falcon Neuro on the ISS [38], near-nadir pass



each pixel on its own



all views, one scene (ESAR [44])

Dissertation: Figure 2.10, Eq. 2.47, Proposition 2.4.1; orbital data and reconstructions: ESAR [44], Falcon Neuro [38]; synthetic aperture [97].

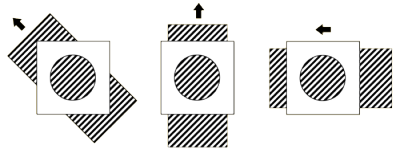
Three different motions look identical through a small window

Through the windows



The visible stripe motions look the same.

Reveal the full patterns



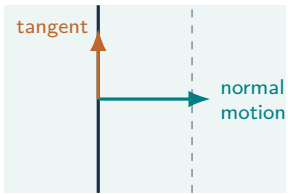
The patterns translate in different directions.

Publisher animations, unchanged: Wolfe et al. [108], Activity 8.2, Movies 5–6.

Dissertation: Proposition 3.3.4 and the following aperture discussion.

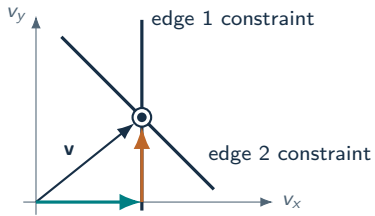
One edge fixes its normal speed; a second orientation fixes the rest

Image: one moving edge



dashed: the same edge a moment later;
events fire as it crosses each pixel

Velocity: what the events allow



normal: fixed by edge 1;
tangent: fixed by edge 2, if it shares the motion

Intersection of constraints [Adelson & Movshon 1982]. Dissertation: Proposition 3.3.4; requirement in Theorem 3.6.1.

Three representations, one quadratic variational problem

$$E_{\mathbf{B},b}(\mathbf{v}) = \underbrace{\frac{1}{2}\int_\Omega (\mathbf{v}^\top \mathbf{B}\mathbf{v} - 2b^\top \mathbf{v})}_{\text{directions the edges measure}} + \underbrace{\frac{\alpha^2}{2}\int_\Omega |\nabla \mathbf{v}|^2}_{\text{neighbors move alike}} \quad .$$

Representation	$\mathbf{B}$	b	Edge measurement from
Time surface	$\nabla\mathcal{T}\nabla\mathcal{T}^\top$	$\nabla\mathcal{T}$	crossing times
Voxel projection	$\nabla a \nabla a^\top$	$-a_s \nabla a$	transported event activity
Structure tensor	$\widetilde{\mathbf{J}}$	$-\mathbf{j}_t$	gradients pooled over a window

Dissertation: Theorem 3.6.1 and its three corollaries; data-plus-smoothness form [16].

Integrated directional information gives a unique flow in H^1

Unified well-posedness theorem

Let $\Omega \subset \mathbb{R}^2$ be bounded, connected, and Lipschitz; $\alpha > 0$. Assume

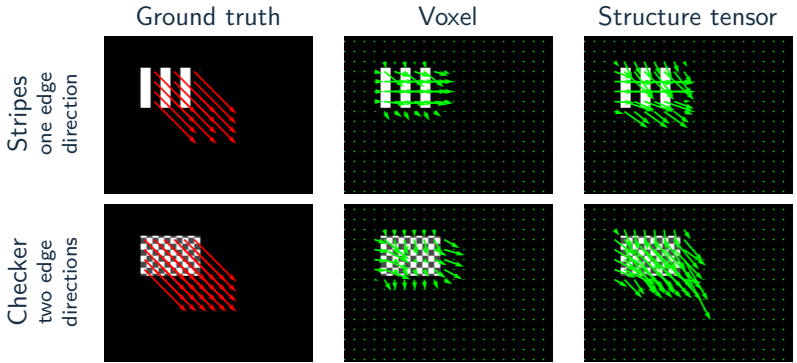
$$\mathbf{B} \in L^\infty, \quad \mathbf{B} = \mathbf{B}^\top \succeq 0 \text{ a.e.}, \quad b \in L^2, \qquad \boxed{\int_\Omega \mathbf{B}(\mathbf{x})\, d\mathbf{x} \succ 0}.$$

Then $E_{\mathbf{B},b}$ has a unique minimizer in $H^1(\Omega;\mathbb{R}^2)$.

Nonparallel edges control constant velocities; smoothness supplies motion where $\mathbf{B} = 0$.

Dissertation: Theorem 3.6.1, Eq. 3.59; classical quadratic-flow theory [56].

Two edge directions resolve more of the motion than one

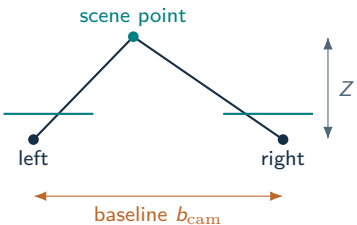


Dissertation: Figure 3.4; v2e [45] simulation, same window and arrow scale.

Stereo matches temporal profiles by encouraging agreement

$$\mathcal{J}_d(\mathbf{f}_L, \mathbf{f}_R) = \sum_{c \in \{L,R\}} \underbrace{\left(\tfrac{1}{2} \left\| \mathbf{A}\mathbf{f}_c - \mathbf{y}_c \right\|^2_{\mathbf{W}_c} + \tfrac{1}{2} \left\| \mathbf{\Lambda}\mathbf{f}_c \right\|^2 \right)}_{\text{each camera fits its own events}} + \underbrace{\tfrac{\gamma}{2} \left\| \mathbf{f}_L - \mathbf{f}_R \right\|^2}_{\text{encourages agreement}}$$

$\mathbf{y}_L$ at pixel (x, y), $\mathbf{y}_R$ at $(x - d, y)$; $\|\mathbf{v}\|^2_{\mathbf{W}} = \mathbf{v}^\top \mathbf{W}\mathbf{v}$, weights updated between solves



$$Z = \frac{f_{\text{cam}} b_{\text{cam}}}{d}, \qquad d = x_L - x_R > 0.$$

For fixed d and frozen weights, $\mathcal{J}_d$ has a unique minimizer.

Dissertation: Eq. 4.24, Section 4.5, Eq. 4.11, Theorem 4.6.3. Geometry is schematic.

WLS scores the most points, 2.66M, at competitive depth error

Method	Mean depth error (cm) ↓ FI1	FI2	FI3	FI4	log RMSE ×100 ↓	Scored points (M)
WLS (ours)	73.83	**58.35**	70.33	59.24	**44.54**	**2.66**
Piatkowska adaptation [77]	**67.96**	58.75	**68.88**	**54.63**	44.72	1.20



MVSEC [35] indoor flying 1–4; each method scored on its own estimates in $[0.5, 8]$ m.
Three more methods in the backup table.

Dissertation: Table A.5; MVSEC [35].

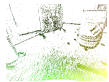
Each next step starts from a result in the dissertation

Sensor modeling	Learned weights	Shared scenes	Edge hardware
error terms (slide 10)	a smooth solution map	the null space of B_{SAR}	the tridiagonal structure
Measure them on real sensors; bound the estimate error.	Learn $\mathbf{\Lambda}$ in an outer problem [109].	Find motion-invariant modes; apply to orbital streams [44].	Time and power on a Raspberry Pi 5; update only active pixels.

Dissertation: Chapter 5, Future Directions; physical connection (p. 24); Proposition 2.4.1. External paper cited above.

What the data constrain and what the models prove

Each estimate is unique once the model supplies what events leave free.

	 **Reconstruction**	 **Optical flow**	 **Stereo depth**
Events constrain	brightness increments	motion directions	paired temporal features
Model proves	**Unique, stable history** fixed positive weights	**Unique H^1 flow** integrated identifiability	**Unique inner solve** fixed disparity and weights

Dissertation: Theorem 2.2.5, Theorem 3.6.1, Theorem 4.6.3; Table 5.1.

Thank you.

Must every temporal penalty entry be positive?

For arbitrary $\mathbf{\Lambda}$, the exact algebraic criterion is

$$\mathbf{A}^\top \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda} \succ 0 \quad \Longleftrightarrow \quad \ker \mathbf{A} \cap \ker \mathbf{\Lambda} = \{0\}.$$

Why?

The quadratic form vanishes precisely when both $\mathbf{Az} = 0$ and $\mathbf{\Lambda z} = 0$.

On a connected first-difference chain, $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$.

- A single nonzero diagonal penalty entry removes that constant mode.
- Fixing one reference value or imposing zero mean also removes it.

Takeaway. Any choice of reference level resolves the mathematical ambiguity; a measured reference would add physical information.

Dissertation: Lemma 2.2.4; discussion following Lemma 4.4.1.

Kernel performance depends on the scene and adaptive weights

	MVSEC indoor		MVSEC outdoor	
Kernel	PSNR	CN	PSNR	CN
B-spline	8.66	259.87	7.29	258.68
Delta	8.49	261.98	7.28	267.99
Gaussian	8.68	259.79	7.23	259.42
Rectangular	8.58	261.10	7.31	263.33

- The temporal operator stays the same; represented data and adaptive penalties change.
- Conditioning and reconstruction bias are separate properties.
- Reference-image scores also measure the baseline the model selects.

Dissertation: Table 2.4, Table 2.5.

What the reconstruction scores measure

Quantity	What it compares	What it establishes
PSNR / SSIM	Reconstruction and reference images	Image agreement under the scoring convention
TC / TG	Temporal changes in the output	Smoothness of the returned sequence
ERC	Magnitudes of reconstructed changes and represented events	Correlation with event activity
CN	Sensitivity of each pixel's system matrix, averaged over pixels	A numerical conditioning diagnostic

A simple check on interpretation

A constant sequence has $\mathrm{TC} = 1$ and $\mathrm{ERC} = 0$ under the stated conventions.

Takeaway. Consider reference fidelity, event consistency, and conditioning together. Smoothness alone rewards a static output.

Dissertation: Evaluation metrics (p. 31); Table 2.3; SSIM [40].

Stable estimates can still have regularization bias

For fixed scalar $\lambda > 0$, $\mathbf{y} = \mathbf{A}\mathbf{f}^\dagger + \boldsymbol{\eta}$, and $\mathbf{K} = \mathbf{A}^\top\mathbf{A} + \lambda^2\mathbf{I}$,

$$\mathbf{f}_\lambda - \mathbf{f}^\dagger = -\lambda^2\mathbf{K}^{-1}\mathbf{f}^\dagger + \mathbf{K}^{-1}\mathbf{A}^\top\boldsymbol{\eta}.$$

The two terms are regularization bias and propagated observation noise.

Takeaway. Stability controls sensitivity to the data; accuracy is a separate property.

Dissertation: Theorem 2.2.5, Proposition 2.2.7; direct bias-noise identity.

The physical event stream loses more than one baseline

The binned increment matrix has $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$, but the ideal sensor can lose time-varying signals too.

$$L_1(t) = L_0, \qquad L_2(t) = L_0 + a\sin(2\pi t/T), \qquad 0 < a < c_{\mathrm{th}}, \quad t \in [0, T].$$

Both start at L_0, emit no events, and have positive intensities $I_j(t) = e^{L_j(t)}$.

Takeaway. The chain nullspace describes the discrete increment surrogate; the physical sensor adds sub-threshold ambiguity.

Dissertation: Eq. 2.1, Lemma 2.2.4; direct ideal-sensor example.

Frozen weights give a unique, stable inner solve

Fix d, $\mathbf{W}_L, \mathbf{W}_R \succ 0$, $\mathbf{\Lambda} \succ 0$, and $\gamma \geq 0$. Define

$$\mathbf{K}_c = \mathbf{A}^\top \mathbf{W}_c \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda}, \qquad \mathbf{M}_W = \begin{pmatrix} \mathbf{K}_L + \gamma \mathbf{I} & -\gamma \mathbf{I} \\ -\gamma \mathbf{I} & \mathbf{K}_R + \gamma \mathbf{I} \end{pmatrix}.$$

For $\mathbf{z} = (\mathbf{z}_L, \mathbf{z}_R)$,

$$\mathbf{z}^\top \mathbf{M}_W \mathbf{z} = \sum_{c=L,R} \left(\left\| \mathbf{W}_c^{1/2} \mathbf{A} \mathbf{z}_c \right\|^2 + \left\| \mathbf{\Lambda} \mathbf{z}_c \right\|^2 \right) + \gamma \left\| \mathbf{z}_L - \mathbf{z}_R \right\|^2.$$

Conclusion

$\mathbf{M}_W \succeq \lambda_{\min}^2 \mathbf{I} \succ 0$: the frozen quadratic has exactly one global minimizer.

Dissertation: Eq. 4.34, Theorem 4.6.3, Theorem 4.6.11.

Motion-induced events and the aperture problem

For a straight edge under steady illumination, let $\mathbf{n}$ be its unit normal, $\mathbf{t}_\perp$ a unit tangent, and s its normal speed:

$$L(\mathbf{x}, t) = \ell(\mathbf{n} \cdot \mathbf{x} - st), \qquad \nabla L = \ell' \mathbf{n}, \qquad L_t = -s\ell'.$$

Where $\ell' \neq 0$, brightness constancy gives

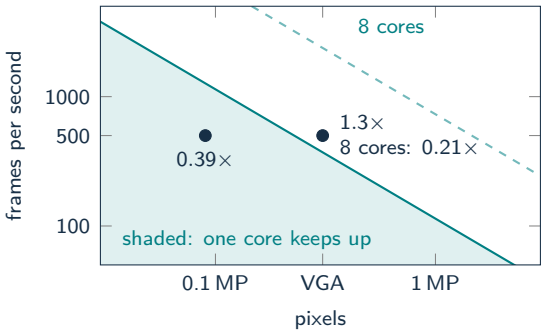
$$L_t + \nabla L \cdot \mathbf{v} = 0 \iff \mathbf{n} \cdot \mathbf{v} = s \iff \boxed{\mathbf{v} = s\mathbf{n} + \tau \mathbf{t}_\perp, \quad \tau \in \mathbb{R}.}$$

- Every such constant velocity gives the same local pattern and ideal events from the same reset reference.
- For $s = 0$, pixel values stay constant in the ideal edge interior; endpoints lie outside the observation window.

Takeaway. The edge determines normal speed; its tangential velocity remains free. Illumination changes can also trigger events under the sensor threshold rule.

Dissertation: Proposition 3.3.4 and following aperture remark; Eq. 2.1.

One CPU core streams about 114 million pixel updates per second, with 16 bytes of state per pixel



$$\text{pixels} \times \text{frames/s} \;\leq\; 1.14 \times 10^8 \text{ per core}$$

8.8 ns per pixel update; 16 B of state per pixel

Batch work follows scene activity: skipping pixels without events cuts it 10× at 6.8% active (0.43 → 0.04)

Points: two recordings, a frame every 2 ms, float32; labels give processing time ÷ recording time. 8-core line: measured at VGA.

Dissertation: Timings: edge implementation, October 2026; Thomas algorithm [32].

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